FARM MANAGEMENT SYSTEM

Platform	Salesforce Lightning / Developer Edition
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Project Workspace	Skill Wallet — Farm Management System

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Project Overview

The **Farm Management System** is a Salesforce-based application designed to organize and manage agricultural information in a centralized system. The project uses Salesforce custom objects, relationships, Lightning tabs, reports, dashboards, and security configuration to represent the main entities involved in farm operations.

The supplied screenshots show a Salesforce Lightning application named **Farm Management** with navigation items for **Farms, Farmers, Crops, Buyers, Reports,** and **Dashboards**. The project is being configured in a Salesforce Developer Edition environment.

The system focuses on maintaining structured information about farmers and their farms, crop-related information, buyers, and management reports. The reporting configuration also demonstrates how Salesforce can convert stored records into grouped reports and dashboard visualizations.

Project Scope

- Maintain farmer information and associate farms with farmers.
- Maintain farm information such as farm name and location.
- Maintain crop records for agricultural activities.
- Maintain buyer records for crop-related business activities.
- Create reports from the configured Salesforce objects.
- Create dashboard components to visualize report data.
- Use Salesforce security features such as profiles, roles, users, and sharing rules.

Objectives

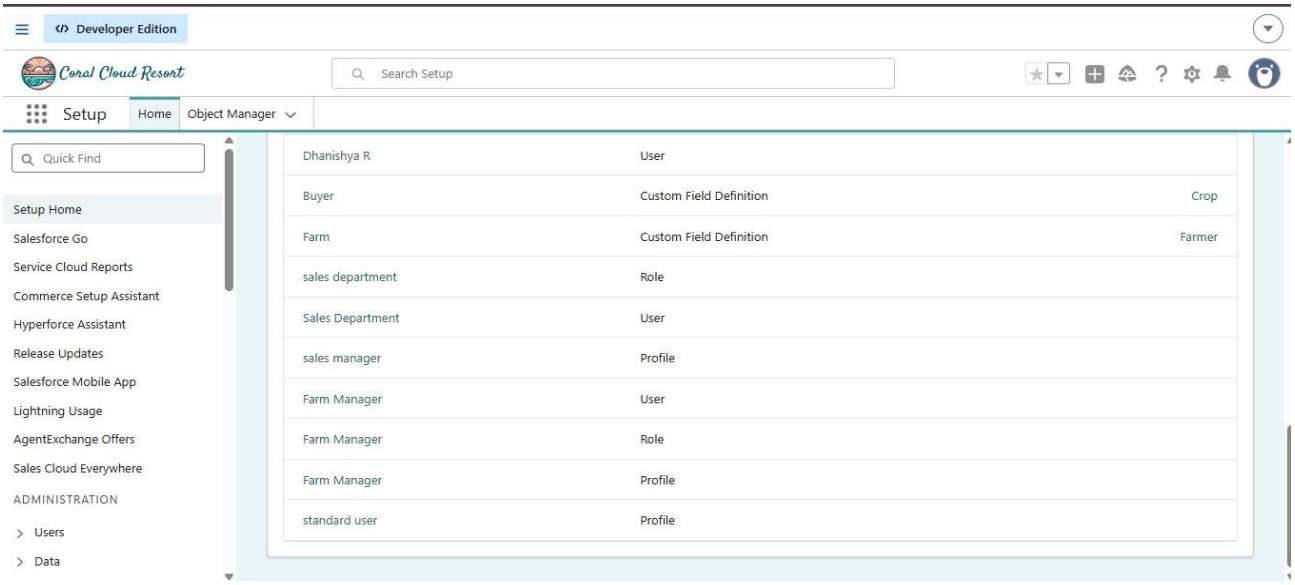
The main objectives of the Farm Management System are:

- Centralize agricultural information in one Salesforce application.
- Reduce dependence on scattered spreadsheets or manually maintained records.
- Create relationships between important farm-management entities.
- Make farmer and farm information easier to search, view, and report.
- Provide reports for management analysis.

- Provide dashboards for quick visual understanding of report results.
- Apply Salesforce access-control concepts to manage who can view or modify records.
- Build practical experience with Salesforce administration and configuration.

Expected Benefits

A centralized CRM-style system can provide a consistent structure for agricultural data, improve visibility of records, simplify reporting, and support controlled access to information. Because the project is built on Salesforce, many administrative functions can be configured declaratively without developing a separate traditional web application.



Technology and Salesforce Components

Component	Use in the Project
Salesforce Lightning	Application interface, navigation, records, reports and dashboards.
Salesforce Developer Edition	Development and configuration environment visible in the screenshots.
Custom Objects	Farmers, Farms, Crops and Buyers represent the main project data entities.
Fields & Relationships	Store attributes and connect related records.
Custom Tabs	Expose project objects such as Farm in the Lightning navigation.
Reports	Create structured views such as Farmer with Farms and crop/buyer reporting.

Dashboards	Visualize report information through charts and components.
Profiles / Roles / Users	Control user access and organizational hierarchy.
Sharing Rules	Extend record access between users, roles or groups according to project requirements.

Functional Modules

Farmer Management

The Farmer module stores information about farmers. The report evidence shows fields such as Farmer Name, Contact Number, District, State, Village, and owner-related fields. Farmer records are used as the parent/grouping information in the Farmer with Farms report.

Farm Management

The Farm module stores farm records. The demonstrated report contains Farm Name and Farm Location and displays these values under the corresponding farmer.

Crop Management

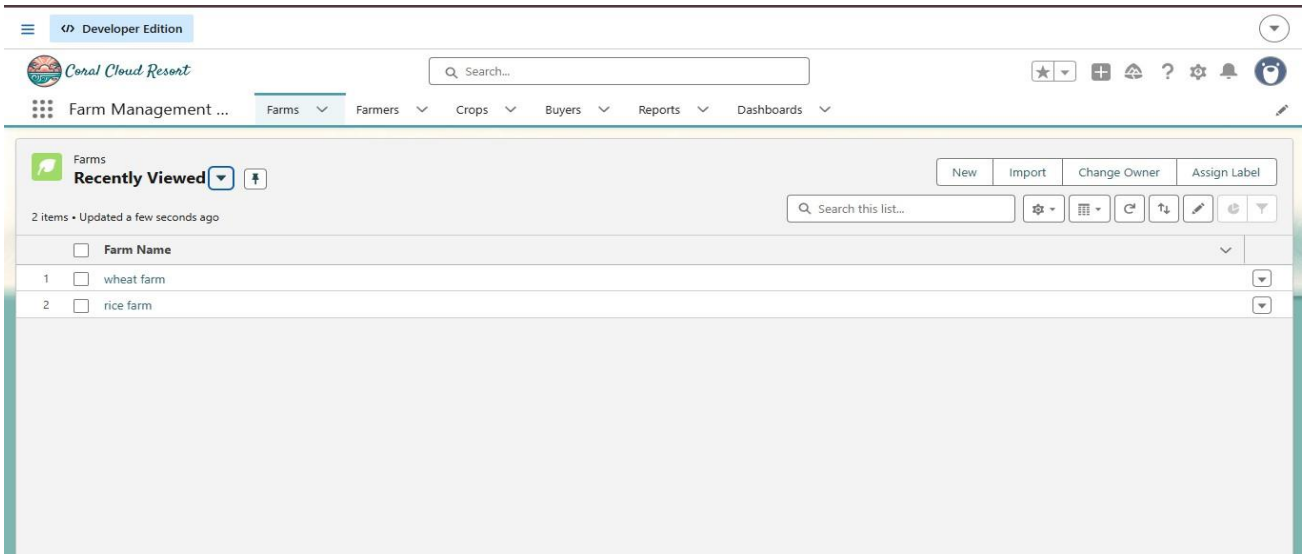
The Crops object is exposed as a navigation tab in the Farm Management application. The supplied report-builder screenshot shows available crop report types including Crops, Crops with Farm, Activities with Crops, and Crop History.

Buyer Management

The Buyers object is also exposed as a navigation tab. The Skill Wallet activity requires a report using Crops and Buyers; the supplied screenshot shows that a direct standard report type named Crops with Buyers / Buyers with Crops was not available in the report-type picker at that point.

Reporting and Dashboard

Reports are used to organize records into columns and groups. Dashboards then present selected report information visually for quick analysis.



Data Model and Relationships

The clearest relationship demonstrated by the supplied evidence is the **Farmer–Farm** relationship. A farmer can be associated with farm records, and the Farmer with Farms report groups farm records under the farmer name.

Illustrative relationship

Parent / Main Entity	Related Entity	Purpose
Farmer	Farm	Connect each farm to its farmer and allow farmer-based reporting.
Crop	Farm	Crop reporting can use the available Crops with Farm relationship/report type shown in Salesforce.
Crop	Buyer	Required by the Skill Wallet activity for crop/buyer reporting; a standard combined report type was not visible in the supplied report picker screenshot.

Reports and Analytics

Farmer with Farms Report

A report named **Farmer with Farms** was created and saved in the **Public Reports** folder. The report contains Farmer Name, Farm Name, and Farm Location. The records shown in the supplied evidence include farmers such as Jayam and Ram, with associated farm names and locations.

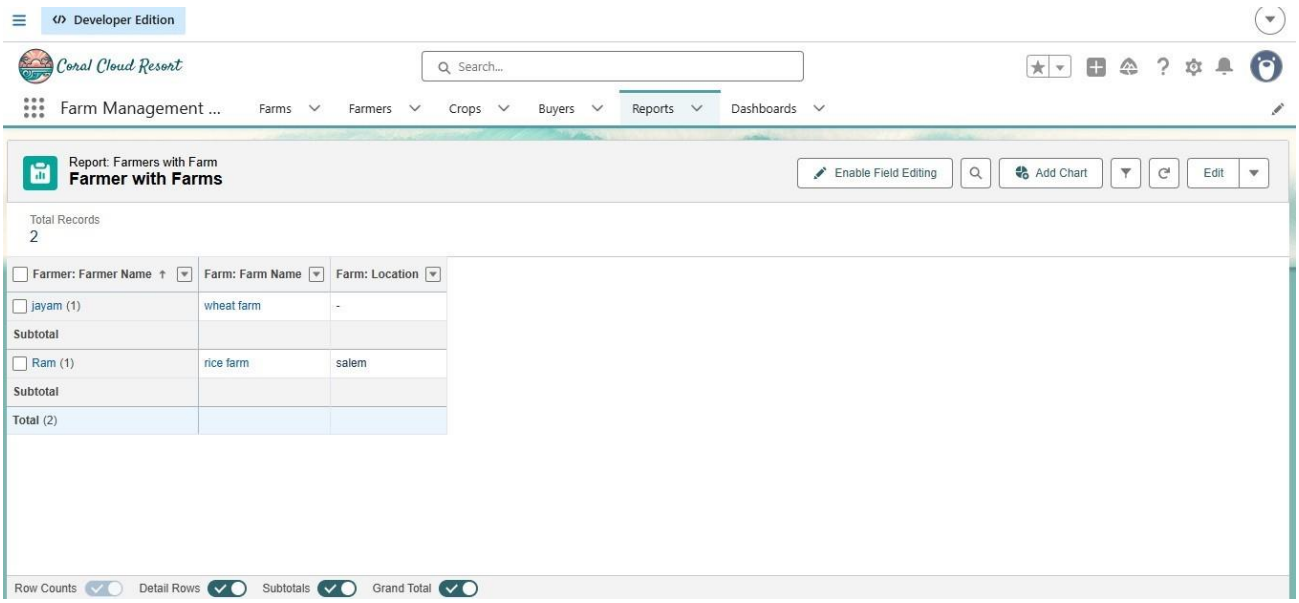
The report is grouped by **Farmer: Farmer Name**. This produces a structure where each farmer is shown as a group and the related farm records appear beneath that group.

Crop / Buyer Reporting Activity

The Skill Wallet activity asks for a report using the Crops and Buyers objects. In the supplied report-type picker, searching for “crop” displayed Activities with Crops, Crops, Crops with Farm, and Crop History, but no Crops with Buyers or Buyers with Crops report type. This is an important configuration point for completing the activity.

Report Design Process

- Open App Launcher and search for Reports.
- Select New Report.
- Choose the required report type.
- Start the report.
- Use Outline → Columns to select fields.
- Use Group Rows to organize records by a meaningful field.
- Save the report in the required folder, such as Public Reports.



Dashboard

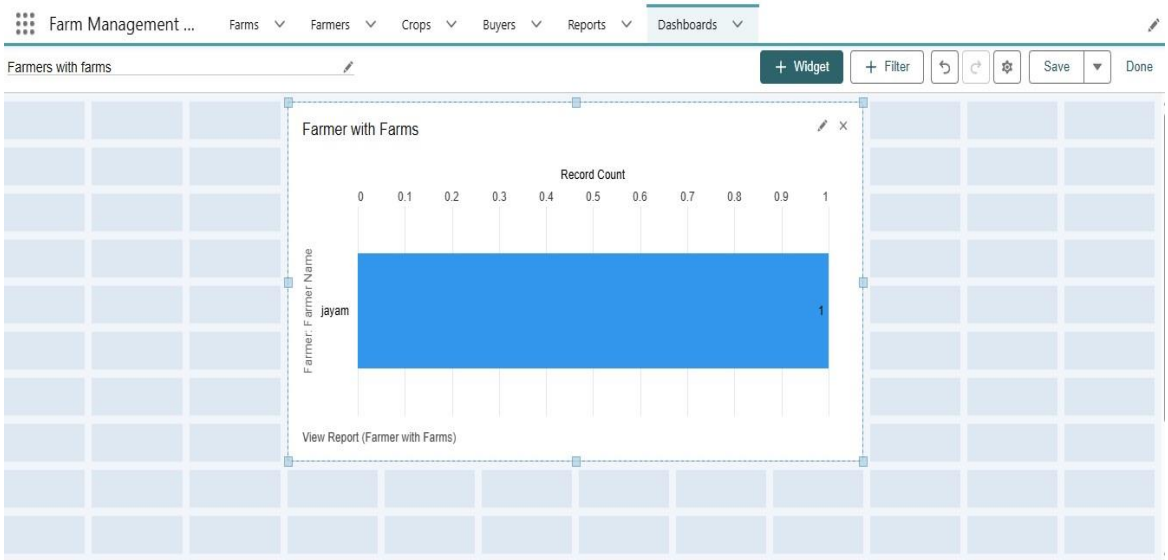
A dashboard named **Farmers with farms** is shown in the supplied evidence. It contains a chart component based on the Farmer with Farms report.

The displayed chart uses **Record Count** and groups the visualization by **Farmer: Farmer Name**. The example dashboard shows a bar for a farmer and a count value, demonstrating how the report can be transformed into a visual summary.

Dashboard workflow

- Create and save the source report.
- Open Dashboards and create/open the required dashboard.
- Add a Widget/Chart.
- Select the Farmer with Farms report as the source.
- Choose a chart type and display the grouped record count.
- Save the dashboard.

The dashboard provides a quick management view instead of requiring users to read every individual report row.



Tabs and Object Configuration

The Farm Management Lightning application navigation shown in the screenshots contains the following major items:

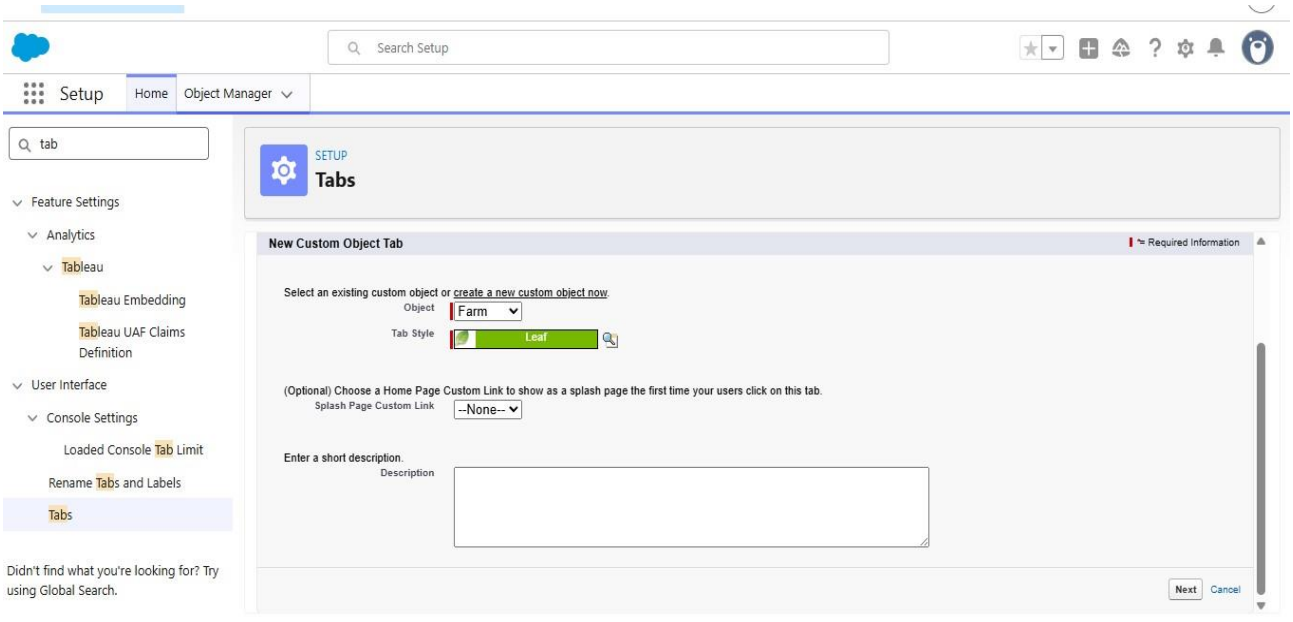
- Farms
- Farmers
- Crops
- Buyers
- Reports
- Dashboards

A Salesforce custom object tab is also demonstrated for the Farm object. The tab configuration screen shows the Farm object selected and a tab style icon chosen. This allows the object to be accessed as a navigation item in the application.

Object-level configuration

The supplied object settings screenshot shows the Farm object with the following options enabled:

- Allow Reports
- Allow Activities
- Track Field History
- Allow Sharing
- Allow Bulk API Access
- Allow Streaming API Access
- Deployment Status: Deployed



Security and Access Management

The Skill Wallet project includes security-related activities covering profiles, roles, users, and sharing rules. These features are important because farm information may need to be visible to some users while restricted from others.

Profiles

Profiles define baseline permissions for users, including access to objects and system capabilities. The project progress screen shows a Profile activity as part of the Farm Management System configuration.

Roles

Roles represent the organization's record-access hierarchy. The project includes a Role activity and therefore uses Salesforce role configuration as part of its security model.

Users

Users are assigned appropriate profiles and roles so that they can work with the application according to their responsibilities. The project progress screen includes a Users activity.

Sharing Rules

Sharing rules can extend record access to appropriate roles or groups. The Skill Wallet activities include a sharing-rule task involving Buyer records and Sales department access with Read/Write permission. The exact recipient should match the role/group configured in the Salesforce org rather than being replaced by an unrelated role.

Conclusion

The Farm Management System demonstrates how Salesforce can be configured as an agriculturefocused information-management application. The project brings together custom objects, object relationships, Lightning navigation, reports, dashboards, and security concepts in one environment.

The supplied evidence confirms a working Farmer with Farms reporting flow and a dashboard visualization based on that report. It also shows the Farm Management application structure and object configuration. Remaining configuration activities can be completed progressively, with particular attention to the Crop–Buyer reporting requirement and the project's security activities.